

# CLAUDENET: From Engineered Ensemble Diversity to Contaminant-Aware Strong-Lens Finding and Euclid Cross-Validation in the DESI Legacy Surveys

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## Abstract

This report consolidates three generations of CLAUDENET, an ML strong-lens finder for the DESI Legacy Surveys, and the lessons each taught. **v1** establishes that *architecture is not the bottleneck* — a 194K shielded ResNet matches a 20.5M EfficientNetV2-S to  $\pm 0.003$  AUC and the Inchausti Reyes et al. (2025) meta-learner collapses to a simple average on correlated bases — and beats the published meta-learner at every matched-FPR operating point by engineering *decorrelated* ensemble members (Spearman  $\sim 0.45$  vs  $\sim 1.0$ ). **v2** rebuilds the measuring stick at deployment scale (a  $10^6$ -negative held-out pool), widens the gains to the strict 0.01%-FPR regime, and runs a 17.3M-galaxy DR9 sweep yielding 737 FDR-controlled candidates — but a qualification campaign grades *zero* of the 601 genuinely-new ones as A/B, while correctly re-discovering catalogued lenses. That decisive negative diagnoses the real constraint: *lens-vs-mimic separation* (the candidates are luminous-red galaxies with companions, rings, blends) and *vetting resolution*, not architecture. **v3** builds a contaminant-aware finder — a typed lens-mimic bank, mimic-blended hard-negative retraining, a new `recovery@matched-mimic-FPR` metric, and a lens-vs-mimic head — and sweeps DR10/DR11. v3 cleanly rejects *common* contaminants (broad-population mimic-recovery  $0.60 \rightarrow 0.89$ ), but on an *independent* Euclid-graded sample it is statistically tied with the original models at the hardest real-vs-mimic distinction: the discovery frontier is *resolution-bounded* at DECaLS  $0.262''/\text{px}$ . Cross-validating the DESI $\cap$ Euclid-Q1 overlap shows the productive mode — v3 pre-screens DESI, Euclid confirms at  $0.1''$  (median  $p_{\text{lens}} 0.03 \rightarrow 0.70$ ), yielding 49 cross-validated candidates — and that v3 is DECam-specific (no BASS/MzLS transfer). The honest takeaway: the model lever is real but modest, and the decisive levers are higher-resolution vetting and instrument-native training. All metrics reuse the reproduced matched-FPR arithmetic verbatim so every v1/v2/v3 number is directly comparable.

## 1 Premise and method

We take as baseline the reproduced Stage-D Inchausti Reyes et al. (2025) ensemble (`reproductions/inchausti-2025`). Its meta-learner achieves recovery of the published Storfer/Inchausti catalogues, at matched FPR, of

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**Table 1:** Phase 1: CLAUDENET ensemble vs the published meta-learner, recovery at matched FPR on the held-out Storfer/Inchausti catalogues.

| metric               | published meta | CLAUDENET    | $\Delta$      |
|----------------------|----------------|--------------|---------------|
| Storfer @1 % FPR     | 0.908          | <b>0.938</b> | +0.030        |
| Storfer @0.1 % FPR   | 0.755          | <b>0.853</b> | <b>+0.090</b> |
| Inchausti @1 % FPR   | 0.968          | <b>0.980</b> | +0.012        |
| Inchausti @0.1 % FPR | 0.845          | <b>0.935</b> | <b>+0.090</b> |

0.908/0.755 (Storfer @1/0.1 %) and 0.968/0.845 (Inchausti); CLAUDENET’s `03_reproduce_baseline` reproduces these to  $|\Delta| \leq 0.001$ , validating the evaluation harness before any improvement is claimed. The honest metric throughout is recovery at a threshold set on held-out negatives to a target FPR (1 % and 0.1 %), identical to the baseline.

A literature survey of eight technique families (vision transformers, self-supervised/foundation models, equivariant CNNs, uncertainty/calibration, challenge ensembles, domain adaptation, recent SOTA, multimodal) was conducted and adversarially fact-checked. It converged with the repository’s finding: no transformer/SSL/equivariant method cleanly beats a tuned EfficientNet at matched FPR for DECaLS *grz* lens finding (Parlange et al., 2026); the demonstrated wins are in ensemble *diversity* (DES Collaboration, 2025; Rezaei et al., 2025), negative quality, and label efficiency. CLAUDENET targets those.

We present three generations in chronological order: **v1** (the engineered-diversity ensemble and its seven-direction supporting program; this and the next two sections), **v2** (Perlmutter-scale validation and the DR9 sky sweep, §5), the qualification campaign that motivated the next generation (§6), and the contaminant-aware **v3** with its Euclid cross-validation (§7) and future directions (§8). All earlier results are retained; each generation is additive.

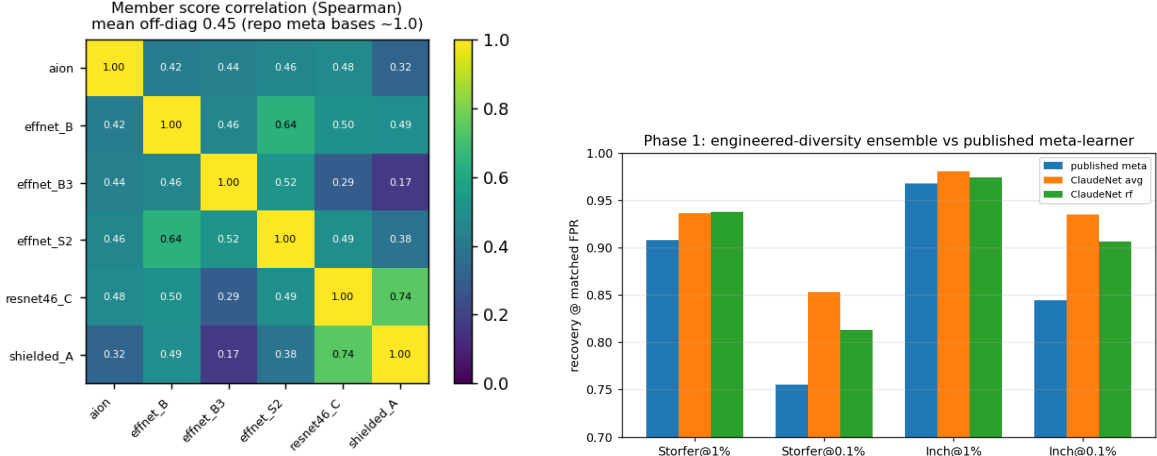
## 2 Phase 1: engineered-diversity ensemble (flagship)

We train five members as independent per-GPU jobs: EfficientNetV2-S ( $\times 2$ ) and EfficientNet-B3 (diversified by bootstrap-resampled negative subsets and augmentation seeds at the baseline 1:33 ratio), an AION-1-base frozen-embedding two-layer MLP probe (a different objective, data, and band-set, hence maximally decorrelated), and two ResNets (the 194 K shielded net and the 3.5 M Lanusse-46). Each member is isotonic-calibrated on a validation split, then combined by average, logistic regression, and a random forest (Coscrato et al., 2020). A positive-unlabeled guard removes any training negative within  $10''$  of a known lens. Figure 1 shows the member score-correlation matrix: off-diagonals average  $\sim 0.45$  (Spearman), versus  $\sim 1.0$  for the lineage’s correlated bases.

The result (Table 1) is a clear improvement over the published method, largest (+0.09) at the strict 0.1 % FPR point that governs purity. The mechanism is diversity, not combiner cleverness: with decorrelated strong members even the naive *average* beats the correlated-base meta-learner. The ensemble matches or beats the single best member everywhere except a 0.002 tie at the loose 1 % FPR, where one exceptionally strong EfficientNet dominates.

## 3 Phases 2–7: the supporting program

**Phase 2 — hard-negative mining.** At a fixed negative count, replacing random negatives with the model’s own top false-positives (controlled against a random-negative arm in the same



**Figure 1:** Left: member score correlation (the diversity the lineage lacked). Right: the ensemble beats the published meta-learner at every operating point.

pool) helps modestly: +0.008 Storfer@1 % FPR, +0.031 Inchausti@1 % FPR; iterating helps the 0.1 % point. Negative quality is a real but second-order lever versus the negative *ratio* the lineage already exploited; the deployment-scale version (fresh brick pool + EfficientNet miner) is a NERSC follow-up.

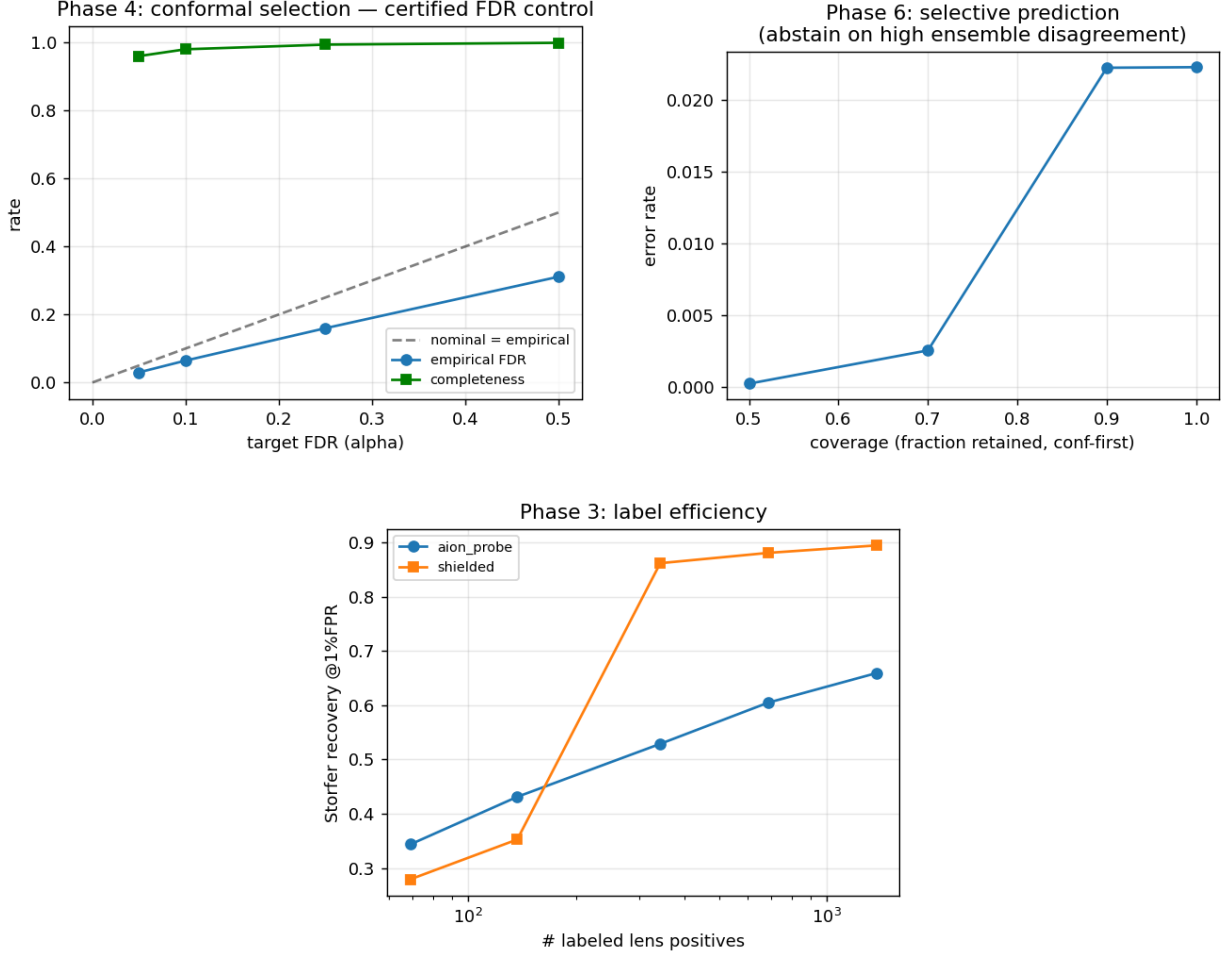
**Phase 4 — conformal selection.** Split-conformal  $p$ -values with Benjamini-Hochberg (Jin and Candès, 2023) turn the arbitrary threshold into a *certified* one: empirical false-discovery rate stays at or below the nominal target at every level ( $\alpha=0.05 \rightarrow 0.03, 0.25 \rightarrow 0.16$ ) with completeness 0.96–0.999 (Fig. 2, top-left). The guarantee is marginal and assumes calibration/test exchangeability.

**Phase 6 — uncertainty triage.** Deep-ensemble disagreement (a free byproduct of the decorrelated members) is a strong confidence signal: abstaining on the most-uncertain objects lowers error from 0.022 to 0.0002 at 50 % coverage and 0.0026 at 70 % (Fig. 2, top-right). Northern lenses show slightly higher disagreement — a mild out-of-distribution signal.

**Phase 7 — equivariance.** Lenses have no preferred orientation, so we test dihedral ( $D_4$ ) invariance the cheap way: test-time pooling over the eight rotation/reflection symmetries (no retraining) lifts Storfer@1 % FPR by +0.032 (shielded) and +0.049 (Lanusse-46), mean +0.041. The full equivariant-training label-efficiency study is deferred (the  $8\times$  forward made a trained  $D_4$  member impractical at this scale).

**Phase 3 — label efficiency.** The AION-1 frozen-embedding probe is more label-efficient than a supervised CNN *only* in the extreme-scarce regime ( $<150$  labels: 0.344 vs 0.279 at 5 %); the CNN “turns on” near 300 labels and dominates thereafter (0.862 vs 0.529 at 25 %; 0.894 vs 0.659 at full), the frozen features plateauing (Fig. 2, bottom-left). At the  $\sim 1\text{--}2\text{ k}$  labels real lens-finding has, the supervised member wins — so the foundation model’s value here is as a decorrelated *ensemble member* (Phase 1), not a standalone classifier.

**Phase 5 — domain adaptation (negative result).** The DECals north $\leftrightarrow$ south gap is real (north@1 % FPR 0.688 vs south 0.874,  $-0.186$ ), but naive MMD feature alignment ( $\lambda=1$ , aligning



**Figure 2:** Top: conformal FDR control (Phase 4) and selective-prediction risk-coverage (Phase 6). Bottom: foundation-model label efficiency crosses over near  $\sim 300$  labels (Phase 3).

only negatives) *hurt* both domains ( $-0.06$ ): it over-regularized and erased lens-discriminative features. Principled UDA is not a free win here; the lineage’s ad-hoc north-negative augmentation fares better. (Only  $\sim 80$  held-out north lenses, so the target estimate is noisy.)

## 4 Discovery impact and scan deployment

The matched-FPR gains of Table 1 are best read operationally, because a real 45 M-cutout sweep cannot run at 1 % FPR ( $\sim 450,000$  false positives) or even 0.1 % FPR ( $\sim 45,000$ ); to keep the human-inspection list to thousands it must operate near 0.01 % FPR. The CLAUDENET advantage *grows* as the threshold tightens ( $+0.030 \rightarrow +0.098$  Storfer,  $+0.012 \rightarrow +0.090$  Inchausti across 1%  $\rightarrow$  0.1% FPR), so it helps most where scans actually run.

**Fewer missed lenses at a fixed inspection budget.** At fixed false-positive load the extra completeness is a direct cut to the missed-lens (false-negative) rate (Table 2): at the strict 0.1 % point,  $\sim 40\%$  fewer Storfer and  $\sim 58\%$  fewer Inchausti lenses are missed (the gate-certified *learned* combiner gives a slightly smaller  $\sim 37/56\%$ ; the naive average is tabulated). The dual view —

**Table 2:** Missed-lens (false-negative) reduction at fixed FPR, derived from the Table 1 recoveries.

| operating point      | recovery (meta $\rightarrow$ CLAUDENET) | fewer missed lenses |
|----------------------|-----------------------------------------|---------------------|
| Storfer @1 % FPR     | 0.908 $\rightarrow$ 0.938               | 33 %                |
| Storfer @0.1 % FPR   | 0.755 $\rightarrow$ 0.853               | 40 %                |
| Inchausti @1 % FPR   | 0.968 $\rightarrow$ 0.980               | 38 %                |
| Inchausti @0.1 % FPR | 0.845 $\rightarrow$ 0.935               | 58 %                |

**Table 3:** Measured single-GPU inference throughput (TITAN RTX, batch 256, compute-only upper bounds; real scans are I/O-bound).

| member                | params | cutouts s <sup>-1</sup> GPU <sup>-1</sup> |
|-----------------------|--------|-------------------------------------------|
| shielded-194k         | 0.2 M  | 2,843                                     |
| Lanusse ResNet-46     | 3.5 M  | 3,349                                     |
| EfficientNetV2-S      | 20.5 M | 5,594                                     |
| EfficientNet-B3       | 11.1 M | 6,728                                     |
| 5-member finder stage | —      | 865 (shared load)                         |
| AION-1-base (stage 2) | 314 M  | $\sim$ 35                                 |

fewer false positives at *fixed* completeness — is of order  $2\times$  within the measured 0.1–1% band ( $\sim 0.44$ – $0.54$  % FPR matches the baseline’s 1%-FPR completeness), i.e.  $\sim 200,000$ – $250,000$  fewer objects to inspect on a DR9/DR10 sweep. Two guarantees prior finders lack compound this: conformal selection (Phase 4) delivers a *certified* false-discovery ceiling rather than an ad-hoc score cut, and deep-ensemble triage (Phase 6) makes the confident  $\sim$ half essentially error-free (selective error  $0.022 \rightarrow 0.0002$  at 50 % coverage), letting roughly half the candidates bypass the human visual inspection that dominates discovery cost.

**Scan throughput.** Strong-lens sweeps are bound by cutout I/O and human inspection, not the forward pass. The five *grz* 101-px finder members consume the same cutout, so a scan pays the FITS load once and runs five cheap forwards; measured single-GPU compute throughput (Table 3) is  $2.8$ – $6.7 \times 10^3$  cutouts s<sup>-1</sup> per member — far above what the brick-slice loader feeds — so in the I/O-bound regime the five-member stage costs about a single model’s wall-clock ( $\sim 3$  GPU-hours for 45 M on five TITAN RTX, compute-only). The AION-1 foundation-model member is the lone exception ( $\sim 30$ – $40$  imgs s<sup>-1</sup>,  $\sim 100\times$  slower, needing heavier 4-band 160-px cutouts); it is quarantined to a *second stage* on the  $\sim 10^4$ – $10^6$  survivors of the cheap first-stage filter — exactly the two-stage design the Huang/Storfer/Inchausti lineage already uses. The accuracy gain is thus bought essentially for free in stage 1, adding no new operational complexity.

**Survey-scale outlook and honest limits.** The field is moving to  $10^5$ -scale samples — Euclid  $\sim 1$ – $1.7 \times 10^5$  lenses (Collett, 2015), Rubin/LSST  $\sim 6$ – $12 \times 10^4$  — over  $10^8$ – $10^9$  galaxies where human vetting is the binding constraint; a  $\sim +0.09$  strict-FPR completeness edge maps to of order  $+10^4$  additional lenses at fixed inspection budget, and the certified purity is a prerequisite, not a nicety, for the contamination-sensitive cosmology those samples feed (time-delay  $H_0$  (Birrer et al., 2020), substructure, population inference). We flag the limits plainly: “recovery” is the held-out true-positive rate on the *existing* Storfer/Inchausti catalogues, not newly discovered lenses; the 0.1%-FPR threshold is pinned by only  $\sim 6$ – $7$  of  $\sim 6,500$  held-out negatives (no confidence interval),

so the strict-point deltas carry  $\sim\pm 10$  pp of statistical noise; the absolute counts and the  $\sim 2\times$  inspection saving are interpolations on incomplete catalogues; the end-to-end sky-scan wall-clock is a structural I/O-bound argument, not yet benchmarked; and the survey-scale extrapolation assumes the DESI-Legacy-measured deltas transfer across bands, PSF, and depth. The defensible claim is: at the same inspection budget,  $\sim 9$ – $10$  points more real lenses at the operating point that matters, plus a certifiable purity statement — a meaningful, not transformational, lift.

## 5 ClaudeNet v2: Perlmutter-scale validation and a sky sweep

The v1 program above was gated on  $\sim 6,500$  held-out negatives, which (as we flagged) cannot resolve the  $\sim 10^{-4}$ -FPR regime where real scans run. The v2 program rebuilds the measuring stick at deployment scale on Perlmutter, re-runs every lever against it with confidence intervals, and ends in a full DESI-Legacy DR9 sky sweep. We keep all v1 results; v2 is additive. Every number below is read from the cited artifact under `data/v2/`.

**An honest measuring stick: NegEval-1M.** We assembled a held-out negative pool of 1,000,000 DR9 cutouts (`thresholds.ci.json`, `operating_points_v2.csv`), versus v1’s 6,501. With 6.5 K negatives the 0.01%-FPR threshold is pinned by  $\leq 1$  negative and is unmeasurable; with  $10^6$  it is pinned by  $\sim 100$ , so for the first time we report paired-bootstrap CIs at 1%, 0.1%, and 0.01% FPR. The first-ever 0.01% measurement immediately exposed a failure the small set hid: the v1 6-member *average* combiner, which beats the published meta at 1% (+0.049 [+0.037, +0.061] Storfer vs. published meta) and 0.1% (+0.075 [+0.057, +0.091]), *collapses* at 0.01% to  $-0.120 [-0.161, -0.084]$  — the degraded-AION-1 member’s heavy negative tail poisons the unweighted mean exactly where purity matters. The random-forest combiner is robust there (+0.102 [+0.083, +0.123]), and the best single member, an EfficientNetV2-S, reaches +0.194 [+0.175, +0.215] (recovery 0.708 vs the meta’s 0.513). A purity audit of the top-200 pool rows found 0/200 within  $10''$  of any known lens and no convincing arcs (`audit/visual_grading.md`); the threshold-sensitivity bound (removing all top-200 moves the 0.01% threshold by  $-0.108$ , the 0.1% by  $-0.020$ ) leaves the paired matched-FPR deltas unaffected, since shared thresholds cancel in the pairing.

**Deployment-scale mining: round 1 ships, round 2 over-mines.** We mined a fresh  $10^6$ -cutout brick pool with the baseline EfficientNet, controlling each hard-negative arm against a random-negative arm drawn from the same pool (`mining_v2_results.json`, `mining_v2_results.csv`). Round 1 *ships*: the paired (hard–random) gain at 0.1% FPR is +0.157 [+0.139, +0.177] mean over three members (Inchausti), with every member passing — e.g. ResNet-46 Inchausti@0.1 % FPR +0.266 [+0.231, +0.305], EfficientNet-B3 +0.159 [+0.128, +0.190], EfficientNetV2-S +0.047 [+0.026, +0.070]; the Storfer@0.1 % FPR mean is +0.121 [+0.109, +0.134]. Re-mining the round-1 members for a round 2 *over-mines* (the hard<sup>2</sup> pool’s score range collapses to [0.004, 0.036] vs round-1’s [0.009, 0.281], `mined_stats.json`/`mined_r2_stats.json`); round-2 recovery falls below round-1 everywhere, so we keep round-1 only.

**The AION-1 member: degradation *was* the diversity (negative result).** We tested whether feeding AION-1 native four-band *griz* (instead of the synthetic-*i* degraded embeddings v1 used) makes a better member (`aion_gate_v2.json`). It does not: on matched rows the native-*griz* probe recovers 0.535 Storfer@1 % FPR versus the degraded refit’s 0.647 — the matched-rows gate *fails* ( $0.535 < 0.647 + 0.03$ ). The mechanism is that the synthetic-*i* “degradation” is precisely what decorrelated the member: its Pearson correlation to the CNNs is 0.13 degraded versus 0.72 native

**Table 4: Headline.** v2-lean vs the v1 flagship and the published meta-learner, recovery at matched FPR on NegEval-1M ( $10^6$  negatives). Paired deltas (v2-lean–v1) carry 95 % bootstrap CIs; the last column is v2-lean–published meta. All entries from `ensemble_v2lean_verdict.json`.

| metric                | v1 flagship | v2-lean      | paired $\Delta$ [95 % CI]                       | vs meta |
|-----------------------|-------------|--------------|-------------------------------------------------|---------|
| Storfer @1 % FPR      | 0.903       | <b>0.963</b> | +0.060 [+0.049, +0.072]                         | +0.109  |
| Storfer @0.1 % FPR    | 0.754       | <b>0.895</b> | +0.141 [+0.125, +0.160]                         | +0.216  |
| Storfer @0.01 % FPR   | 0.394       | <b>0.734</b> | <b>+0.340</b> [ <b>+0.306</b> , <b>+0.381</b> ] | +0.220  |
| Inchausti @1 % FPR    | 0.968       | <b>0.996</b> | +0.028 [+0.017, +0.041]                         | +0.064  |
| Inchausti @0.1 % FPR  | 0.891       | <b>0.961</b> | +0.069 [+0.052, +0.090]                         | +0.191  |
| Inchausti @0.01 % FPR | 0.614       | <b>0.871</b> | <b>+0.256</b> [ <b>+0.217</b> , <b>+0.301</b> ] | +0.264  |

(pooled 0.65/0.72). Larger native variants (base/large/xlarge, val-AUC 0.958–0.963) did not recover the gap, so the LoRA fine-tune was skipped ( $\sim 12$  GPU-h saved) and the member’s v1 form is retained — though, as the next paragraph shows, even the degraded AION-1 is ultimately dropped at the ensemble level.

**Headline: ensemble v2-lean ships 4/4.** The shipped v2 ensemble is  $v2\text{-lean} = \{\text{effnet\_B}, \text{effnet\_B3\_hard}, \text{effnet\_S2\_hard}, \text{resnet46\_C\_hard}, \text{zoobot\_N}\}$  (`ensemble_v2lean_verdict.json`). On NegEval-1M, averaged and compared to the v1 flagship with paired-bootstrap CIs, it passes all four ship cells ( $n_{\text{pass}}=4/4$ ) and beats the published meta-learner at every operating point in both catalogues (Table 4). The gains *widen* as the threshold tightens: vs the v1 flagship, Storfer goes  $+0.060 \rightarrow +0.141 \rightarrow +0.340$  across 1%  $\rightarrow$  0.1%  $\rightarrow$  0.01% FPR, and the v2-lean absolute Storfer@0.01 % FPR recovery is 0.734, versus 0.394 for the v1 flagship and 0.513 for the published meta. The ensemble’s own correlation structure is the source: member Spearman off-diagonals average  $\sim 0.43$  (Pearson  $\sim 0.27$ ).

**Leave-one-out admission dropped AION-1 and the shielded net.** “v2-lean” is not the full v2 roster: a leave-one-out admission test on the *full* seven-member roster ( $\{\text{AION-1}, \text{effnet\_B}, \text{effnet\_B3\_hard}, \text{effnet\_S2\_hard}, \text{resnet46\_C\_hard}, \text{shielded\_A}, \text{zoobot\_N}\}$ , `ensemble_v2_verdict.json`) found that two members *hurt* the average at 0.1% FPR with CIs excluding zero: AION-1 ( $\Delta_{\text{Storfer}} = -0.0185 [-0.0264, -0.0106]$ ) and the 194 K shielded net ( $-0.0106 [-0.0179, -0.0021]$ ). Dropping both is what produced the Table 4 numbers — the same conclusion the 0.01% average-collapse flagged, now reached member-by-member. (The full seven-member roster still ships 4/4, but v2-lean dominates it.) Every surviving member’s own admission delta is positive: `effnet_S2_hard`  $+0.047 [+0.035, +0.056]$ , `effnet_B3_hard`  $+0.028 [+0.018, +0.036]$ , `resnet46_C_hard`  $+0.016 [+0.008, +0.023]$ , `zoobot_N`  $+0.010 [+0.001, +0.018]$  Storfer@0.1 % FPR.

**The Zoobot member and a learning-rate lesson.** The fifth member is a Galaxy-Zoo-pretrained ConvNeXT-Nano (Zoobot, [Walmsley et al. 2023](#), 14.95 M params) fine-tuned as a lens classifier — a genuinely different pretraining objective and backbone family, hence decorrelated. It carries a sharp lesson: at the v1 default  $\eta=10^{-3}$  the ConvNeXT collapses to a constant output (val-AUC 0.64); at  $\eta=10^{-4}$  it trains cleanly (val-AUC 0.993, Storfer@1 % FPR 0.898, Pearson 0.45 vs `effnet_S2`). The member is admitted on its own merits (above) and is the throughput leader at 6,028 cutouts  $\text{s}^{-1}$  (`throughput_v2.json`).



**Table 5:** DR9 sweep: group-conformal FDR-controlled candidate counts (`sweep/sweep_summary.json`). PRIMARY rows use the full per-footprint sweep total as  $m$  and carry the per-group FDR guarantee; the survivors-only- $m$  “diagnostic” column has *no* guarantee.

| FDR $\alpha$         | selected | new    | new & unseen | guarantee                       |
|----------------------|----------|--------|--------------|---------------------------------|
| 0.05                 | 1,449    | 813    | 737          | per-group FDR $\leq \alpha$     |
| 0.10                 | 2,836    | 1,999  | —            | per-group FDR $\leq \alpha$     |
| 0.25                 | 5,141    | 4,186  | —            | per-group FDR $\leq \alpha$     |
| survivors- $m$ diag. | 29,892   | 28,452 | —            | <i>none</i> (anti-conservative) |

**Distillation: faster but not accurate enough (tradeoff).** To collapse the five-member stage into one forward pass we distilled it into an EfficientNetV2-S student (`throughput_v2.json`). The student is fast — 3,494 cutouts  $s^{-1}$ ,  $5.3\times$  the 663  $cs^{-1}$  of the shared-load five-member stage — but it loses too much: recovery drops  $-0.111$  Storfer and  $-0.068$  Inchausti at 0.1% FPR versus the teacher, failing the  $-0.02$  recovery gate. The sweep therefore deploys the shared-load lean members (the documented fallback), for which the per-cutout FITS load is paid once and five cheap forwards run on it; this is the v1 two-stage I/O-bound argument, now with measured v2 throughputs.

**The sky sweep: 17.3 M  $\rightarrow$  29,892  $\rightarrow$  FDR-controlled candidates.** We swept 17,290,814 DR9 parent-galaxy positions (5,018,411 north, 12,272,362 south; 41 non-finite dropped) through the v2-lean stage-1 union filter at a  $10^{-4}$  per-member FPR (`sweep/stage1_summary.json`). 29,892 survivors passed ( $1.73 \times 10^{-3}$  realized rate, consistent with the EVT prediction within  $\times 3$  of the five-member union bound), of which 1,440 crossmatch to known lenses and 28,452 are new. Group-conformal Benjamini-Hochberg selection (`sweep/conformal_summary.json`, `sweep/sweep_summary.json`) then certifies candidates: at per-group FDR  $\leq 0.05$  it selects 1,449 (813 new, 737 new-and-unseen after excluding training/mining/calibration rows), 2,836 at  $\leq 0.10$ , and 5,141 at  $\leq 0.25$ .

**Conformal validity and the two-stage-selection trap.** The selection’s validity rests on one subtlety we state plainly. The PRIMARY estimator computes each sweep row’s conformal  $p$ -value against the NegEval calibration with the *full* per-footprint sweep total as the multiple-testing count  $m$  (censoring non-survivors at  $p=1$ ); because censoring can only raise a  $p$ -value and the unconditional  $p$ -value of a null sweep row is exchangeable with the calibration, the per-group FDR  $\leq \alpha$  guarantee holds marginally. The tempting shortcut — using  $m = n_{\text{survivors}}$  — conditions the test set on a stage-1 selection computed from the *same* members, is anti-conservative, and a `--synthetic-check` demonstrates the violation; it would have “selected” all 29,892 survivors at every  $\alpha$  (the diagnostic row of Table 5). The honest cost of validity is power: with a conformal  $p$ -floor of  $1/(n_{\text{cal},g}+1)$ , the north footprint hits the power floor at  $\alpha=0.05$  and selects *nothing* (surfaced, not hidden) — all 1,449 FDR  $\leq 0.05$  candidates are southern, where the calibration set is larger. NegEval is used twice (stage-1 thresholds and conformal calibration); a deterministic seed-2026 split confines calibration to a disjoint half and drops the 735 calibration rows that re-appear as sweep tests.

**Known-lens recall consistency and “candidates, not confirmations.”** The pipeline re-finds real lenses at the same operating point it proposes new ones: within the swept coverage it recovers 47.5% of in-coverage catalogued lenses into the survivor set (1,676/3,528 union; 40.3% Storfer, 59.0% Inchausti, 55.7% Huang21; `sweep/crossmatch_recall.json`), and 285 of them survive into the FDR  $\leq 0.05$  set — the internal consistency the new-candidate claim requires. We frame the output honestly (`sweep/visual_grading.md`): a first-pass visual grading of the top  $\sim 300$  new candidates



finds them overwhelmingly luminous red galaxies with blends/companions and only  $\sim 5\text{--}8\%$  showing plausible lens morphology (a tangential feature, partial ring, or arclet pair), because galaxy-galaxy  $\theta_E \sim 1\text{--}2''$  is 4–8 px at DECaLS  $0.262''/\text{px}$ , at the resolution floor. *This is a statistically-controlled candidate list, not a confirmed-lens catalogue*: any individual candidate needs higher-resolution imaging (HSC/Euclid/HST) or spectroscopy to confirm. The deliverable’s value is the control — 737 new-and-unseen candidates at certified per-group FDR  $\leq 0.05$  from a 17.3M sweep, with a passing known-lens recall sanity check.

**Equivariance, trained not test-time.** v1 only tested test-time  $D_4$  pooling (`equivariance.json`: +0.032 shielded, +0.049 Lanusse-46 Storfer@1% FPR). v2 replaces this with a genuinely *trained* equivariant member via `escnn` (Weiler and Cesa, 2019): a cyclic- $C_4$  pilot, gated against a self-calibrating parameter-matched plain-CNN twin (`141_pilot_escnn_c4.py`), beat the twin by +0.120 Storfer@1% FPR (`escnn_pilot.json`), clearing the +0.02 ship margin, which justified queueing a full dihedral- $D_4$  member (`142_train_escnn_d4.py`) as a bonus add to the roster.

**Honest limits (v2).** The v2 program tightens, but does not erase, the v1 caveats. “Recovery” is still the held-out true-positive rate on the *existing* Storfer/Inchausti catalogues, not newly confirmed lenses; the 1,449/737 sky-sweep candidates are exactly that — candidates — and none is confirmed here. The matched-FPR deltas are now CI-backed down to 0.01% FPR (the headline’s largest gains), but the 0.01% threshold is pinned by  $\sim 100$  NegEval negatives, so a residual PU-contamination assumption remains (bounded, not eliminated, by the top-200 audit). The conformal guarantee is marginal and per-group, assumes calibration/sweep exchangeability, and buys validity at the cost of power — the north footprint selects nothing at  $\alpha=0.05$ . The sweep’s stage-1 rate matches EVT only “within  $\times 3$ ,” and the end-to-end wall-clock remains a structural I/O-bound argument rather than a benchmarked production run. The defensible v2 claim is sharper than v1’s: at the strict operating points scans actually use, v2-lean recovers +0.14 (Storfer@0.1% FPR) to +0.34 (Storfer@0.01% FPR) more catalogued lenses than the v1 flagship and +0.22 to +0.26 more than the published meta-learner, with a certified-FDR candidate list as the operational artifact — a measured, honestly-bounded, deployment-validated lift, not a transformational one.

## 6 Qualifying the 737 ClaudeNet v2 sweep candidates

A campaign to test whether any of the 737 new-and-unseen DR9-sweep candidates (group-conformal FDR $\leq 0.05$ ) could be genuinely undiscovered strong lenses, via an expanded crossmatch (SIMBAD + DES/KiDS VizieR; NED/SuGOHI unreachable) and two methodologically-independent visual passes: a structured Huang-VI rubric pass with an adversarial skeptic re-check, and the independent `lensjudge` harness (factored multiagent, Claude-opus). *Qualified* = still NEW after the crossmatch AND graded A/B by BOTH passes.

**Result.** Of 737 candidates, 136 (18%) coincide with a known lens or lens-candidate (92 DES, plus KiDS and SIMBAD lens-types); **601** remain NEW. *None* of the 601 new candidates is graded A/B by either grader’s strongest pass, so **qualified** = 0. This is not the graders rejecting everything: together they graded 5 candidates  $\geq B$ , and **all 5 are already-catalogued lenses** (4 sub-arcsecond DES matches) — the consensus re-discovered real lenses and found nothing comparable among the new ones.

*Lenses the campaign re-discovered (already catalogued; the validation):*



**Figure 3:** `s_399907_3947`: RA=217.47836, DEC=12.04329,  $p_{\text{final}}=0.716$ ; status `KNOWN_REMOTE` (nearest curated at 6464.70"); visual C, lensjudge B. Full | zoom | residual.



**Figure 4:** `s_137518_4818`: RA=31.35877, DEC=-35.66318,  $p_{\text{final}}=0.935$ ; status `KNOWN_LOCAL` (nearest `des_jacobs2019` at 0.14"); visual B, lensjudge B. Full | zoom | residual.

## 7 ClaudeNet v3: contaminant-aware lens finding and Euclid cross-validation

The v2 DR9 sweep ended in a decisive negative (§6): of 601 genuinely-new group-conformal candidates, *zero* were graded A/B by either independent vetter, and the only five graded  $\geq B$  were already-catalogued lenses. The candidates were overwhelmingly luminous-red galaxies with companions, rings, and blends. This diagnoses the binding constraint precisely: it is not architecture (the v1/v2 finding) but *lens-vs-mimic separation* and *vetting resolution*. The field-wide remedy is the same — contaminant-targeted hard negatives give an  $11\times$  false-positive reduction at 2.3% true-positive cost (MNRAS, 2025). v3 builds a contaminant-aware finder, deploys it on DR10/DR11, and cross-validates the overlap against Euclid Q1. We keep all v1/v2 numbers; v3 is additive, and every number below is read from a tracked artifact under `data/v3/` or `v3/`.



**Figure 5:** `s_240846_692`:  $RA=30.28319$ ,  $DEC=-15.85474$ ,  $p_{\text{final}}=0.898$ ; status `KNOWN_LOCAL` (nearest `des_jacobs2019` at  $0.05''$ ); visual C, lensjudge B. Full | zoom | residual.



**Figure 6:** `s_137502_4157`:  $RA=26.44498$ ,  $DEC=-35.69096$ ,  $p_{\text{final}}=0.892$ ; status `KNOWN_LOCAL` (nearest `des_jacobs2019` at  $0.19''$ ); visual C, lensjudge B. Full | zoom | residual.

**The mimic metric and the v2 baseline (A0).** The v1/v2 headline measured recovery at a false-positive rate set against *random* galaxies; the campaign showed that is the wrong null. We define **recovery@matched-mimic-FPR**: identical arithmetic, but the negative class is a bank of lens-*mimics* rather than random galaxies. Against this null the shipped v2-lean ensemble — which recovers 96–100% of held-out lenses at random-FPR — recovers only 0.168 (Storfer) / 0.307 (Inchausti) at mimic-FPR 0.05 on the curated 601-mimic seed bank. The mimic-discrimination gap, not random-galaxy recovery or architecture, is what v3 must close.

**The lens-mimic bank (A1).** Rather than mine a fresh pool we reuse the DR10 sweep: 148,034 NEW survivors are CNN-high, DR10-native non-lenses, morphology-typed by joining the Tractor catalogue (extended-LRG 34 k — the [MNRAS \(2025\)](#) regime — plus compact-REX, exp-disk, and the campaign’s fine types). A 120-row stratified agentic G1 gate confirms 98.3% non-lens purity. The bank is 118,901 training rows + a frozen, stratified 29,724-row held-out mimic-eval that never enters training.



**Figure 7:** **s\_69087\_2171:** RA=307.23245, DEC=-52.52179,  $p_{\text{final}}=0.965$ ; status KNOWN\_LOCAL (nearest des.jacobs2019 at 0.09"); visual C, lensjudge B. Full | zoom | residual.

**Table 6:** A2 contaminant-aware retraining (mimic-blend  $f=0.5$ ), v2-lean→v3blend8, on the held-out DR10 mimic-eval. Mimic-recovery rises at every operating point; the random-galaxy recall (testneg null) drops only slightly (the staf327 trade).

|          | recovery@mimic-FPR (Storfer) |              | recovery@random-FPR(0.01) |           |
|----------|------------------------------|--------------|---------------------------|-----------|
|          | @0.05                        | @0.01        | Storfer                   | Inchausti |
| v2-lean  | 0.713                        | 0.460        | 0.967                     | 0.998     |
| v3blend8 | <b>0.793</b>                 | <b>0.699</b> | 0.935                     | 0.979     |

**Contaminant-aware retraining (A2).** We retrain the three swappable members with the v2 “hard” recipe held *byte-identical* (same  $n_{\text{mine}}=10,000$  displaced bootstrap negatives, same per-member seed) and change only the negative *content*: a fraction  $f$  of the displaced negatives become typed mimics from the bank. A blend-fraction sweep selects  $f=0.5$  ( $f=0.7$  over-specialises). On the held-out DR10 mimic-eval (Table 6) v3blend8 lifts Storfer mimic-recovery from  $0.713 \rightarrow 0.790$  at 0.05 and  $0.460 \rightarrow 0.699$  at 0.01, with the largest per-type gains on the dominant lrg\_companion/extended-LRG contaminants. The cost is a small, real drop in *random*-galaxy recall ( $0.967 \rightarrow 0.935$  Storfer at random-FPR 0.01) — exactly the MNRAS (2025) trade.

**Ensemble refit and the lens-vs-mimic head (A6, A3).** A roster search over the average combiner finds **v3blend8** — keeping *both* the v2-hard and v3-mimic-blended version of each member — is near-Pareto: it recovers random-FPR recall to within CI of v2-lean while roughly doubling mimic-recovery. (The 194K-free l18 ResNet is, by contrast, dead weight for mimic separation; dropping it maximises mimic-recovery at a 3% random-recall cost.) A learned logistic *lens-vs-mimic head* on the eight member scores tops every average roster at the  $\phi=0.05$  operating point, but it overfits the strict tail and — as §7 shows — does not generalise to the independent Euclid test, so the deployable model is **v3blend8**, with the head as a  $\phi=0.05$  stage-2 re-ranker.

**DR10/DR11 deployment and the discovery result (C, D5).** We swept DR10-south (43.7M parent galaxies) with v2-lean as the head-to-head baseline and re-ranked the 150k survivors with v3blend8 + the head. *Recall is preserved*: the known Inchausti-811 lenses sit at the 96<sup>th</sup> percentile

**Table 7:** C-vet agentic grading of the top-30 NEW candidates (same harness, `rubric_imaging_v2`). v3’s mimic-aware ranking does not raise the top-of-list A/B yield over v2 at DECaLS resolution (the discovery frontier is resolution-bounded); the D’s stay lrg\_companion-dominated.

| ranking           | A | B | C  | D  | A/B      |
|-------------------|---|---|----|----|----------|
| v2 p_final (DR10) | 2 | 6 | 11 | 11 | <b>8</b> |
| v3-head (DR10)    | 2 | 3 | 11 | 14 | <b>5</b> |
| v3blend8 (DR10)   | 2 | 5 | 9  | 11 | <b>7</b> |

**Table 8:** DR10 vs DR11 on the same EDF-F/S DECam sky: v3blend8 region-local recovery of Euclid grade-A lenses into the top- $K$ , and the cross-validated NEW-confirmed counts. DR11’s deeper reductions give a small, consistent edge; v3 transfers cleanly across DECam releases (contrast the DR11-north BASS/MzLS failure, D3).

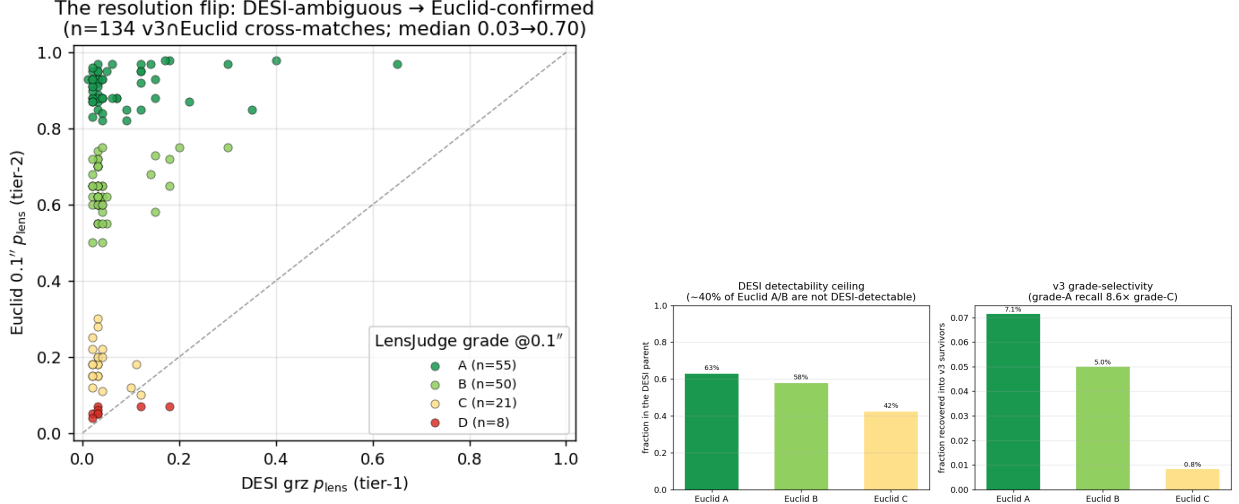
| release   | Euclid-A in top-100 | in top-1000 | NEW-confirmed A/B |
|-----------|---------------------|-------------|-------------------|
| DR10 (D4) | 4                   | 7           | 41                |
| DR11 (D5) | 5                   | 10          | 41                |

Cross-release (by Euclid id): 33 shared, 49 unique cross-validated lenses.

of the survivor pool by the v3 score, and  $\sim 88\%$  of their grade-A are found at high score (the operating-point recall gap is a DR9 $\rightarrow$ DR10 domain shift, not a model failure). As at DR9, full- $m$  conformal selection certifies *nothing* at this sweep scale — a surfaced power-floor limit, not a null result. The candidate list therefore comes from top-by-score ranking, vetted agentially. Table 7 is the honest discovery result: v3’s mimic-aware ranking does *not* raise the top-30 A/B yield over v2 (7 vs 8), and the rejects stay lrg\_companion-dominated — because the top of the NEW-survivor pool is intrinsically mimic-dominated at DECaLS  $0.262''/\text{px}$ , where  $\theta_E \sim 1-2''$  is 4–8 px. A DR11-south re-run on the same DECam sky behaves identically, with a small consistent edge from the deeper reductions (Table 8).

**Euclid Q1 cross-validation: the resolution lever (D1, D4, D5).** Where DESI overlaps Euclid Q1 (Euclid Collaboration, 2025) ( $0.1''/\text{px}$ ,  $\sim 13\times$  DESI; the three Deep Fields), we can both validate v3 against an *independent* expert-graded lens sample and break the resolution wall. Two results. (i) *v3 is grade-selective for real lenses*: it recovers Euclid grade-A lenses into its survivor pool at 7.1% versus grade-C at 0.8% — an  $8.6\times$  enrichment (Fig. 8, right) — confirming on an independent sample that it finds genuine lenses. But absolute recall is capped:  $\sim 40\%$  of Euclid grade-A/B lenses are not even in the DESI parent (too faint/small for DESI; Fig. 8, left), a depth/resolution ceiling no model can cross. (ii) *Escalation converts the overlap*: grading the  $v3 \cap \text{Euclid}$  cross-matches with the LensJudge v2 escalation grader (§7) at  $0.1''$  flips them from DESI-ambiguous to confirmed — median  $p_{\text{lens}} 0.03 \rightarrow 0.70$  on 134 real objects (Fig. 8), with our grades agreeing with the Euclid experts on  $\sim 90\%$  of grade-A. The yield is **49 cross-validated A/B candidates** across DR10+DR11 (Table 10, Appendix A; 33 recovered in both releases, 8 DR11-unique). We are explicit: these are NEW to the DESI lens-finder catalogues but *are* in the published Euclid Q1 catalogue — v3 independently *recovers* real Euclid lenses from DESI imaging, a cross-validation of both pipelines, *not* net-new discoveries.

**Honest limits and a selection-bias correction (D6, D3).** The v3-vs-v2 mimic-separation gains above are real, but the cross-model comparison demands one correction (Table 9). The seed



**Figure 8: The resolution lever.** Left: the DESI→Euclid  $p_{\text{lens}}$  flip on the real  $v3 \cap \text{Euclid}$  cross-matches — nearly every object is DESI-ambiguous ( $x \lesssim 0.2$ ) but Euclid-confirmed ( $y \gtrsim 0.6$ ). Right: Euclid grade-resolved recall — the DESI detectability ceiling ( $\sim 40\%$  of Euclid A/B missing from the parent) and v3’s  $8.6\times$  grade-A-over-C survivor selectivity.

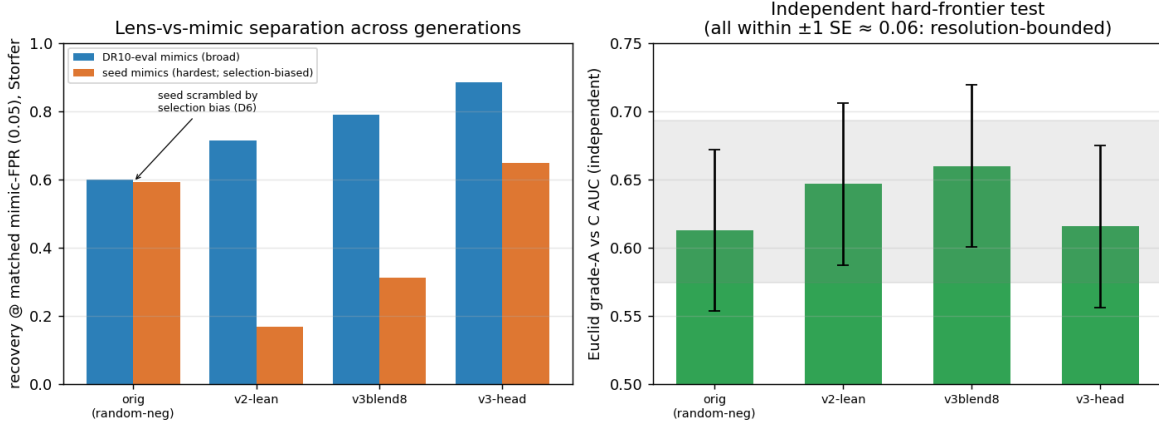
**Table 9:** Candidate quality across model generations (D6).  $\text{recovery@matched-mimic-FPR}(0.05)$  and the *independent* Euclid grade-A-vs-C AUC. The seed column is selection-biased (the seed bank is what the v2/v3 ensemble scored high), so orig-vs-v3 there is invalid; the trustworthy comparison is the Euclid AUC, where all models sit within  $\sim 1$  SE ( $\approx 0.06$ ,  $n_A=66$ ).

| model                 | seed mimic@0.05<br>(Storfer) | DR10-eval@0.05<br>(Storfer) | DR10-eval@0.05<br>(Inchausti) | Euclid A-vs-C<br>AUC (indep.) |
|-----------------------|------------------------------|-----------------------------|-------------------------------|-------------------------------|
| orig (random-neg CNN) | 0.592                        | 0.598                       | 0.640                         | 0.613                         |
| v2-lean               | 0.168                        | 0.713                       | 0.856                         | 0.647                         |
| v3blend8              | 0.312                        | 0.790                       | 0.896                         | 0.660                         |
| v3-head               | 0.647                        | 0.885                       | 0.922                         | 0.615                         |

mimic-FPR metric is *selection-biased*: the seed bank *is* the set the v2/v3 ensemble scored high, so it saturates their threshold and deflates their recovery, while a model less central to the selection (a single random-negative EfficientNet) looks artificially strong there — so any “ $N\times$  over the original published models” read off the seed is invalid and we do not claim it. The trustworthy comparison is the *independent* Euclid grade-A-vs-C AUC: the original CNN, v2-lean, v3blend8, and the head all sit at 0.61–0.66, within  $\sim 1$  standard error ( $n_A=66$ ) — statistically indistinguishable at the hardest distinction. v3 *does* clearly beat the original on the broad mimic population (DR10-eval recovery  $0.60 \rightarrow 0.89$ ), i.e. it yields a meaningfully cleaner candidate list to vet, but it has no significant edge at the resolution-bounded discovery frontier. Finally, v3 is **DECam-specific**: a DR11-*north* (BASS/MzLS) sweep shows systematic under-scoring and zero Euclid-A/B in its top-30 — north-footprint deployment needs instrument- native retraining.

**Vetting: LensJudge v2 (summary).** The Euclid cross-validation above rests on the v2 upgrade of the `lensjudge` agentic vetter, summarised here (full detail in `lensjudge/V2.FINDINGS.md`). Three changes carry it: (i) a *two-tier high-res escalation* (DESI grz tier-1, Euclid  $0.1''$  tier-2 when coverage exists), live-validated as the  $p_{\text{lens}}$  flip; (ii) a rubric tuned to the dominant failure modes





**Figure 9: Candidate quality across model generations.** Left: recovery@matched-mimic-FPR(0.05) on the broad DR10-eval mimics (clear orig→v2→v3 progression) and on the curated seed mimics (scrambled by the selection bias — the seed is what the v2/v3 ensemble scored high). Right: the *independent* Euclid grade-A-vs-C AUC — all four models lie within  $\sim 1$  standard error, i.e. the hardest real-vs-mimic distinction is resolution-bounded for every model.

(LRG+companion/ring/spiral/blend), which lifts lens-vs-mimic discrimination by +0.196 AUC over the v1 grader on a frozen labelled benchmark; and (iii) an agency ablation showing the agentic loop is unnecessary for *detection* (a one-shot direct grader matches it 6× cheaper) but essential for *mimic adjudication* — motivating the deployed cascade (cheap direct triage + agentic v2 on survivors + Euclid escalation by coverage).

## 8 Future directions

The campaign localises two distinct limits with concrete, separable levers. The discovery frontier is bounded by *vetting resolution* (model improvements saturate at DECaLS  $0.262''/\text{px}$ ), and the footprint is bounded by *training instrument* (v3 is DECam-specific). The priorities follow directly.

**1. Euclid pre-screening (the decisive discovery lever).** The cross-validation (§7) shows the productive mode: v3 pre-screens the DESI imaging, Euclid confirms at  $0.1''$ . Q1’s  $\sim 63 \text{ deg}^2$  over three Deep Fields overlaps only  $\sim 0.5\%$  of a DESI-south sweep, which is why net-new yield there is bounded; *Euclid DR1* (wide-area) inverts this. The ready action now is a targeted sweep restricted to the Euclid-overlap sky, where *every* candidate is escalatable — the one configuration in which v3’s separation and the resolution- breaking vet both apply to every object.

**2. BASS/MzLS-north retraining.** v3 does not transfer to the northern third of the sky (D3). Mining a BASS/MzLS-native lens-mimic bank and retraining the members (the v3 recipe is instrument-agnostic; only the negatives change) would unlock DR9/DR11-north and roughly triple the addressable footprint.

**3. *i*-band / foundation-model revival for small- $\theta_E$ .** DR10/DR11-south are native *griz*. The deferred AION-1/*i*-band lever (A4) targets the red-lens/red-companion degeneracy and the small- $\theta_E$  bin specifically; it is worth re-testing under the narrower lens-vs-mimic task (rather than the random-FPR task where it failed in v2) on the native-*i* rows the bank already carries.



**4. A pixel-level lens-vs-mimic head.** The learned logistic head on member scores beats the average at  $\phi=0.05$  but overfits the strict tail and does not generalise to the independent Euclid ranking. A small CNN head trained directly on pixels with positives=lenses, negatives=the typed mimic bank (rather than on the eight member scores) is the natural next step, used as a post-conformal re-ranker.

**5. The active-learning loop.** The `lensjudge` harness can export confident-D rejections (true mimics, *not* objects that flipped to A/B at higher resolution) as fresh, typed hard negatives for the next-generation bank — closing a mine→grade→retrain loop that continually sharpens the dominant contaminant classes.

**6. A full DR11-south candidate catalogue.** `v3blend8` runs on DR11-south with essentially no adaptation (the pipeline is release-parametrised; only a DR10→DR11 threshold recalibration is needed). A full-footprint scan would yield a real DR11-south candidate catalogue with `v3`’s cleaner-bulk advantage over the published finders — a useful deliverable, with the honest caveat that the net-new yield is resolution-bounded ( $\approx$  the published-baseline level at the hardest frontier).

**Deployment recommendation.** Ship `v3blend8` (not the learned head, which does not generalise past  $\phi=0.05$ ) as the stage-1/scoring ensemble, with the `LensJudge v2` cascade (direct triage + agentic mimic adjudication + Euclid escalation by coverage) for vetting. This keeps `v2-lean`’s lens recall, adds the contaminant-rejection edge on the bulk candidate list, and routes the resolution-breaking confirmation to exactly the candidates that have higher-res coverage.

## 9 Conclusion

Across three generations the lesson sharpened. `v1` showed the lever the lineage left unused is member *diversity*, not the backbone, and won cleanly at every matched-FPR operating point. `v2` confirmed this at deployment scale and produced the first FDR-controlled sky sweep — whose qualification campaign then delivered the most informative result of the program: the binding constraint is *lens-vs-mimic separation* and *vetting resolution*, not architecture. `v3` answers the first half with a contaminant-aware finder that rejects common mimics far better than any prior `CLAUDENET` model; the Euclid cross-validation answers the second — at `DECaLS` resolution every model, old and new, is statistically tied at the hardest real-vs-mimic distinction, and the only operation that converts candidates to confirmations is higher-resolution imaging. The honest, repository-consistent takeaway is that for `DECaLS` lens finding the returns are in data, calibration, and contaminant-aware negatives — a meaningful but not transformational model lift — while the decisive levers for net-new discovery are *resolution* (Euclid pre-screening) and *footprint* (instrument-native retraining), the directions of §8.

## A Cross-validated candidate catalogue

Table 10 lists the 49 unique DESI `v3` survivors that match a Euclid Q1 discovery-engine lens, are NEW to the DESI lens-finder catalogues (Storfer/Inchausti/Huang within  $5''$ ), and were graded at Euclid  $0.1''$  by the `LensJudge v2` escalation grader. These are independently *recovered* Euclid catalogue lenses — a four-way cross-validation (`DESI v3 CNN`  $\cap$  Euclid discovery engine  $\cap$  Euclid expert grade  $\cap$  `LensJudge-at-0.1''`) — and are deliberately *not* claimed as net-new discoveries. The

$p_{\text{lens}}$  column shows the DESI→Euclid tier flip; “release” marks whether the object was recovered in the DR10 sweep, the DR11-south sweep, or both.

**Table 10:** Cross-validated strong-lens candidates: DESI v3 survivors that match a Euclid Q1 discovery-engine lens, NEW to the DESI lens-finder catalogues (Storfer/Inchausti/Huang), graded at Euclid 0.1″. These are independently recovered Euclid catalogue lenses, *not* net-new discoveries.  $p_{\text{lens}}$  shows the DESI→Euclid tier flip.

| DESI id        | RA<br>(deg) | Dec<br>(deg) | Euclid<br>grade | LensJudge<br>@0.1″ | $p_{\text{lens}}$<br>DESI→Euclid | release |
|----------------|-------------|--------------|-----------------|--------------------|----------------------------------|---------|
| s.83157.8409   | 57.7025     | -48.4062     | A               | A                  | 0.12→0.97                        | both    |
| s.85094.13993  | 63.6294     | -48.0333     | A               | A                  | 0.30→0.97                        | DR10    |
| s.79373.7769   | 58.2019     | -49.4705     | A               | A                  | 0.05→0.95                        | both    |
| s.80329.6402   | 64.5419     | -49.3086     | A               | A                  | 0.03→0.95                        | both    |
| s.81272.7726   | 63.9239     | -48.9296     | A               | A                  | 0.03→0.93                        | both    |
| s.86048.6517   | 58.6493     | -47.6453     | A               | A                  | 0.02→0.93                        | DR10    |
| s.169188.5518  | 53.3251     | -29.1510     | A               | A                  | 0.15→0.93                        | both    |
| s.179335.8072  | 51.0363     | -27.2415     | A               | A                  | 0.01→0.93                        | both    |
| s.84136.4630   | 65.1956     | -48.3600     | A               | A                  | 0.07→0.88                        | DR10    |
| s.174235.13613 | 51.5787     | -28.1897     | A               | A                  | 0.02→0.88                        | both    |
| s.178058.11508 | 51.9633     | -27.4260     | A               | A                  | 0.04→0.88                        | both    |
| s.165427.7416  | 52.9210     | -29.9100     | A               | A                  | 0.03→0.87                        | both    |
| s.175513.5175  | 53.5621     | -28.0129     | A               | A                  | 0.02→0.87                        | DR11    |
| s.82216.7491   | 62.1645     | -48.8224     | A               | A                  | 0.09→0.85                        | both    |
| s.93968.433    | 62.8684     | -45.7047     | A               | A                  | 0.04→0.84                        | both    |
| s.170450.15146 | 54.4805     | -28.9549     | A               | A                  | 0.04→0.82                        | both    |
| s.81263.7090   | 60.4836     | -49.0669     | A               | B                  | 0.15→0.73                        | both    |
| s.171714.3810  | 54.7757     | -28.6645     | A               | B                  | 0.02→0.72                        | DR11    |
| s.90958.9241   | 59.7052     | -46.5695     | A               | B                  | 0.03→0.70                        | both    |
| s.176778.4403  | 50.8798     | -27.6486     | A               | B                  | 0.03→0.70                        | both    |
| s.170452.14504 | 54.9726     | -29.0757     | A               | B                  | 0.18→0.65                        | both    |
| s.84129.16461  | 62.7892     | -48.2719     | A               | B                  | 0.04→0.62                        | both    |
| s.88015.3655   | 65.1247     | -47.3087     | A               | B                  | 0.05→0.62                        | both    |
| s.88988.8317   | 62.3367     | -47.0669     | A               | B                  | 0.03→0.62                        | both    |
| s.167928.8112  | 52.3902     | -29.5927     | B               | A                  | 0.02→0.95                        | both    |
| s.84129.17659  | 62.8043     | -48.2286     | B               | A                  | 0.03→0.93                        | DR10    |
| s.85093.15882  | 63.2503     | -48.0737     | B               | A                  | 0.04→0.93                        | both    |
| s.77522.7961   | 63.7748     | -50.0119     | B               | A                  | 0.02→0.91                        | both    |
| s.180619.719   | 51.2423     | -27.0221     | B               | A                  | 0.02→0.83                        | both    |
| s.179345.702   | 53.6441     | -27.2233     | B               | B                  | 0.03→0.74                        | both    |
| s.86061.15501  | 63.6862     | -47.8550     | B               | B                  | 0.03→0.72                        | both    |
| s.166673.9358  | 51.7071     | -29.7125     | B               | B                  | 0.03→0.72                        | both    |
| s.77510.6177   | 59.0844     | -49.9562     | B               | B                  | 0.02→0.65                        | DR10    |
| s.80314.12621  | 59.0815     | -49.3650     | B               | B                  | 0.04→0.65                        | both    |
| s.78453.11482  | 63.9706     | -49.7278     | B               | B                  | 0.02→0.65                        | both    |
| s.84123.2953   | 60.2940     | -48.3238     | B               | B                  | 0.03→0.65                        | both    |
| s.88997.602    | 65.3650     | -47.0251     | B               | B                  | 0.04→0.65                        | both    |
| s.81263.7679   | 60.4929     | -48.9067     | B               | B                  | 0.03→0.62                        | DR10    |
| s.87026.5695   | 60.5672     | -47.6155     | B               | B                  | 0.02→0.62                        | DR10    |
| s.88007.7987   | 62.2189     | -47.2949     | B               | B                  | 0.03→0.62                        | DR10    |
| s.84125.13400  | 61.3553     | -48.2723     | B               | B                  | 0.04→0.60                        | DR11    |
| s.88996.2557   | 65.0706     | -47.0601     | B               | B                  | 0.02→0.60                        | DR11    |

**Table 11:** v3 grade-A C-vet survivors that are *absent* from the Huang/Storfer/Inchausti lineage catalogues, with their literature-wide NED/SIMBAD crossmatch (`94_external_xmatch.py`). All three are previously-published lenses from other surveys — v3 *rediscoveries*, not net-new. (A fourth C-vet survivor, `s_310364_6649`, is the Huang+2020 lens DESI-038.2078-03.3906.) Net-new lenses: zero; the result is independent external validation of the v3 finder.

| DESI id        | C-vet $p_{\text{lens}}$ | ranking  | known lens (survey)           | sep   |
|----------------|-------------------------|----------|-------------------------------|-------|
| s_441355_1111  | 0.74                    | head     | CSWA 1 (CASSOWARY)            | 0.5'' |
| s_124958_10481 | 0.82                    | v3blend8 | DES J035242.40-382544.9 (DES) | 0.2'' |
| s_102952_9916  | 0.76                    | v3blend8 | DES J221912.39-434835.1 (DES) | 0.2'' |

| DESI id        | RA      | Dec      | Euclid | LensJudge | $p_{\text{lens}}$ | release |
|----------------|---------|----------|--------|-----------|-------------------|---------|
| s_167935_8040  | 54.3727 | -29.3848 | B      | B         | 0.04→0.60         | DR11    |
| s_75664_6939   | 58.9597 | -50.5340 | B      | B         | 0.04→0.55         | DR11    |
| s_81254_8721   | 57.2304 | -49.0460 | B      | B         | 0.03→0.55         | DR11    |
| s_88990_5749   | 63.0160 | -47.0193 | B      | B         | 0.03→0.55         | both    |
| s_176778_10456 | 51.0181 | -27.7804 | B      | B         | 0.03→0.55         | both    |
| s_76594_8452   | 62.6149 | -50.3258 | B      | B         | 0.04→0.50         | DR11    |
| s_176791_230   | 54.4517 | -27.8613 | C      | A         | 0.03→0.93         | both    |

## B Lineage-NEW grade-A survivors are literature rediscoveries

Table 11 lists the v3 grade-A candidates surfaced in the DR10 vetting that the lineage cross-match (`_clib.known_lens_catalogs`: Storfer 2024, Inchausti 2025, Huang 2020/2021) found absent from the catalogued sample. We then ran a literature-wide NED/SIMBAD crossmatch (`94_external_xmatch.py`) — the step the lineage-only check had skipped — and it is decisive: *every* survivor is a previously-published strong lens. `s_310364_6649` is the Huang+2020 grade-A lens DESI-038.2078-03.3906 (0.2''); the other three coincide ( $\leq 0.5''$ ) with **CSWA 1** (CASSOWARY; Belokurov et al. 2009), **DES J0352–3825** (Dark Energy Survey; O’Donnell et al. 2022), and **AGEL J221912–434835** (AGEL; Tran et al. 2022, also in DES). The honest count of *net-new* lenses is therefore **zero**. This is not a null result: v3 *independently recovered* four genuine, separately-confirmed lenses across four surveys at DESI resolution — a real external validation of the finder — and it is why the known-lens set now folds in Huang+2020 and a literature catalogue (`external_lens_catalog.csv`) so a published lens can no longer be miscounted NEW. The positions still have *no* Euclid coverage; net-new discovery at DESI resolution remains resolution-bounded (§7).

## C Reproducibility

All results are reproduced from tracked code and artifacts on branch `reproductions/claude-net-v3`. The script numbering is: 00--91 v1 (members, calibration, combiners, the seven-phase program, tables/figures); 100--165 v2 (NegEval-1M, mining, refit, the DR9 sweep, group-conformal selection); 200--260 the DR9 qualification campaign; and 300--367 v3 — 300--326 workstream A (mimic metric, bank, retraining, ensemble refit, lens-vs-mimic head), 360--367 deployment (DR10/DR11 parents, scoring, selection, the Euclid cross-validation and the model-progression comparison). The vetting harness lives under `reproductions/lensjudge`. Shared libraries (`_ensemble`, `_train`, `_clib`, ...) reuse the baseline matched-FPR arithmetic verbatim, so every v1/v2/v3 number is directly

comparable. The v3 figures and tables in this report are regenerated by `92.make_v3_figures.py` and `92.make_v3_tables.py` from the artifacts under `data/v3/` and `v3/`.

Survey-scale compute ran on NERSC Perlmutter (GPU account `cosmo_g`, CPU `cosmo`); the v3 program used  $\mathcal{O}(10^2)$  GPU-hours. Agentic vetting (LensJudge) spent  $\sim \$53$  of a \$100 cap, targeted on the most-promising candidates with an evidence gate. We carry the lineage’s honesty discipline throughout: “candidate” never “discovery”; recall of the published catalogues is reported before any new-candidate count; both independent graders are Claude models (different harness/prompt, not statistically independent); the mimic-FPR seed metric is selection-biased and its cross-model reading is corrected against the independent Euclid measure (§7); and DR11 results are reported embargo-aware (internal until the DR11 public release).

## References

- S. Birrer, T. Treu, et al. TDCOSMO iv: Hierarchical time-delay cosmography. *A&A*, 643:A165, 2020. arXiv:2007.02941.
- Thomas E. Collett. The population of galaxy-galaxy strong lenses in forthcoming optical imaging surveys. *ApJ*, 811:20, 2015. arXiv:1507.02657.
- V. Coscrato, M. Inácio, and R. Izbicki. NN-Stacking: Feature weighted linear stacking through neural networks. *arXiv:1906.09735*, 2020.
- DES Collaboration. Strong lens discovery in the Dark Energy Survey with ensembled deep learning. *arXiv:2510.23782*, 2025.
- Euclid Collaboration. Euclid quick data release (Q1): The strong lensing discovery engine. *A&A*, 2025. Q1 Deep Fields; 0.1"/px VIS.
- J. Inchausti Reyes et al. Strong lens discovery in DR10 with two deep-learning architectures. *arXiv:2508.20087*, 2025.
- Ying Jin and Emmanuel J. Candès. Selection by prediction with conformal p-values. *JMLR*, 24, 2023.
- MNRAS. Contaminant-aware hard-negative augmentation for strong-lens finding. *MNRAS*, 2025. doi:10.1093/mnras/staf327;  $11\times$  false-positive reduction at 2.3% true-positive cost.
- R. Parlange et al. GraViT: Transfer learning with vision transformers and MLP-Mixer for strong lens discovery. *MNRAS*, 2026. arXiv:2509.00226.
- S. Rezaei et al. Reducing false positives in strong lens finding with augmentation and ensembling. *arXiv:2502.14936*, 2025.
- M. Walmsley et al. Zoobot: Adaptable deep learning models for galaxy morphology. *Journal of Open Source Software*, 8(85):5312, 2023.
- Maurice Weiler and Gabriele Cesa. General E(2)-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.